DATA 412/612 · Statistical Programming in R · Week 14

14.2 Iteration: Loops and purrr

Pass the function in. map() walks a list, across() walks columns, and a for-loop is for the index.

Pass a function in, and let `map()` or `across()` walk the pieces. A for-loop is still the tool when you need the index, or when you do not know how many steps you will take.

By the end of this lecture you should be able to:

1. Tell a vector, a list, and a data frame apart, including `[` and `[[`.
2. Treat a function as an object you can pass to another function.
3. Repeat a function down columns with `across()`.
4. Walk a list or the columns of a data frame with `map()`.
5. Write a named function, or an anonymous one with `\(x)`.
6. Pick `map_dbl()`, `map_int()`, `map_chr()`, or `map_lgl()` from the type you want back.
7. Fit one model per group with `group_modify()` or with `split()` plus `map()`.
8. Walk two or more inputs with `map2()` and `pmap()`.
9. Use `walk()` when the point is a file or a plot, not a returned value.
0. Put a `map()` next to the for-loop it replaces.

Lecture 14.1 has the longer account of vectors and lists. The rowwise lecture has the longer account of `across()`.

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1. A list, then a data frame

An atomic vector holds one type. A list can mix types, including a function. A data frame is a list of columns that all have the same length.

```
x <- list(greeting = "hello", number = 1, operation = mean)
x[1]
x[[1]]
x$greeting
```

```
## $greeting
## [1] "hello"
## [1] "hello"
## [1] "hello"
```

`x[1]` is still a list. `x[[1]]` and `x$greeting` are the element inside.

```
df <- tibble(
  name = c("Alice", "Bob", "Carol"),
  age = c(21, 25, 22),
  score = c(85, 90, 88)
)
df["age"]
df[["age"]]
```

```
## # A tibble: 3 × 1
##     age
##   <dbl>
## 1    21
## 2    25
## 3    22
## [1] 21 25 22
```

`df["age"]` is a one-column data frame. `df[["age"]]` is the numeric vector. `map()` can walk a data frame because each column is one element of that list.

---

2. A function is an object

`mean` is an object. `mean(1:10)` calls it. You can store the function under another name and call that name.

```
f <- mean
f(c(10, 20, 30))
```

```
## [1] 20
```

`map()` and `across()` take the function as an argument. Write `mean`, not `mean()`. The parentheses would call it immediately, before `map()` has an element to hand it.

---

3. A for-loop, pre-allocated

`seq_along(x)` is `1, 2, …, length(x)`. Prefer it to `1:length(x)`, which misbehaves on an empty `x`. Growing a vector with `c()` inside the loop is slow. Make the output first, then fill it.

```
x <- 1:5
cumvec <- vector("double", length(x))
total <- 0
for (i in seq_along(x)) {
  total <- total + x[[i]]
  cumvec[[i]] <- total
}
cumvec
```

```
## [1]  1  3  6 10 15
```

`vector("double", n)` is `n` zeros. You overwrite them. `x[[i]]` pulls one element. `cumvec[[i]] <-` writes one element back.

`while` when `n` is unknown

A `for` loop needs the sequence up front. A `while` loop keeps going until a condition is false:

```
nheads <- 0
flips <- 0
while (nheads < 3) {
  flips <- flips + 1
  if (rbinom(1, size = 1, prob = 0.5) == 1) {
    nheads <- nheads + 1
  }
}
```

Use `while` when you cannot say how many steps you need. Add `flips` as the last line when you want the count. That number changes from run to run.

---

4. `across()` on columns

`across()` is the dplyr form of "the same function, many columns." It lives inside `summarize()` or `mutate()`. The full treatment, including `rowwise()` and `c_across()`, is the rowwise lecture. Two patterns matter here.

One number per column, inside `summarize()`. Pass `mean` and `sd` as functions. `.names` builds names from the column and the function.

```
mtcars |>
  summarize(across(
    c(mpg, wt),
    list(mean = mean, sd = sd),
    .names = "{.col}_{.fn}"
  ))
```

```
##   mpg_mean   mpg_sd wt_mean     wt_sd
## 1 20.09062 6.026948 3.21725 0.9784574
```

`where(is.numeric)` selects by type. `is.numeric` has no parentheses. `where()` wants the function, not the result of calling it.

```
mtcars |>
  summarize(across(where(is.numeric), median))
```

```
##    mpg cyl  disp  hp  drat    wt  qsec vs am gear carb
## 1 19.2   6 196.3 123 3.695 3.325 17.71  0  0    4    2
```

Inside `mutate()`, a function that returns one number is repeated down the column. Every row of `x` below becomes `mean(x)`.

```
xyz <- tibble(x = c(10, 20, 30), y = c(20, 40, 60))
xyz |> mutate(across(x:y, mean))
```

```
## # A tibble: 3 × 2
##       x     y
##   <dbl> <dbl>
## 1    20    40
## 2    20    40
## 3    20    40
```

`summarize(across(...))` collapses to one row per group. `mutate(across(...))` keeps the rows.

---

5. `map()`

`map(.x, .f)` applies `.f` to each element of `.x` and returns a **list**.

```
measurements <- list(
  blood_glucose = c(90, 95, 100),
  age = c(20, 30, 40),
  heart_rate = c(70, 75, 80)
)
map(measurements, mean)
```

```
## $blood_glucose
## [1] 95
##
## $age
## [1] 30
##
## $heart_rate
## [1] 75
```

Each element is handled on its own. The names come along.

A data frame is a list of columns, so `map()` handles one column at a time. `select(where(is.numeric))` drops the factor columns of `iris` first. `sd` of a factor is not a useful number.

```
iris |>
  select(where(is.numeric)) |>
  map(sd)
```

```
## $Sepal.Length
## [1] 0.8280661
##
## $Sepal.Width
## [1] 0.4358663
##
## $Petal.Length
## [1] 1.765298
##
## $Petal.Width
## [1] 0.7622377
```

`map(mtcars, class)` asks for the class of every column. That is a useful first look at a new table. The result is a list because `class()` is allowed to return more than one string. `cut` in `diamonds` is an ordered factor, so `class()` returns `"ordered"` and `"factor"`. `map_chr()` would stop there. Section 8 is the rule.

Extra arguments of `.f` go after the function. They are the same on every element.

```
map_dbl(mtcars, mean, na.rm = TRUE)
```

```
##        mpg        cyl       disp         hp       drat         wt       qsec
##  20.090625   6.187500 230.721875 146.687500   3.596563   3.217250  17.848750
##         vs         am       gear       carb
##   0.437500   0.406250   3.687500   2.812500
```

---

6. Write the function once

Copying a block three times is a good moment to name a function. Calling that function three times is a good moment to use `map()`.

Several Excel files with the same odd layout: skip four rows, take sheet 2, and supply the names. One function holds that layout. This chunk is not run here. There are no workbooks in the notes.

```
library(readxl)

read_weird_excel <- function(file) {
  read_excel(
    file,
    sheet = 2,
    skip = 4,
    col_names = c("id", "value", "group")
  )
}

files <- c("data1.xlsx", "data2.xlsx", "data3.xlsx")
datasets <- map(files, read_weird_excel)
combined <- bind_rows(datasets)
```

`datasets` would be a list of three data frames. `bind_rows()` stacks them.

`keep()` holds the elements where a test is true. `discard()` drops them. Then `map_dbl()` summarizes what remains.

```
iris |>
  keep(is.numeric) |>
  map_dbl(mean)
```

```
## Sepal.Length  Sepal.Width Petal.Length  Petal.Width
##     5.843333     3.057333     3.758000     1.199333
```

---

7. Anonymous functions

A function you need once does not need a name. These four calls all mean "the mean, dropping missing values," and each one prints the same list.

Way	What you type	Name of the current element	Remember
The function's name	`map(x, mean, na.rm = TRUE)`	the function's own first argument	no parentheses on `mean`
`function()`	`function(x) mean(x, na.rm = TRUE)`	the name you choose	works in every version of R
Lambda `\()`	`\(.x) mean(.x, na.rm = TRUE)`	the name you choose. `.x` here is only a name you picked	short form of `function()`, from R 4.1 on. Use this in new code
Tilde `~`	`~ mean(.x, na.rm = TRUE)`	always `.x`. A second argument is always `.y`	the older purrr formula. It still runs

The lambda and the tilde can look alike because both can say `.x`. In `\(.x)` you wrote that name. In `~`, purrr supplies `.x` and you do not write parentheses.

```
map(measurements, mean, na.rm = TRUE)
map(measurements, function(x) mean(x, na.rm = TRUE))
map(measurements, \(.x) mean(.x, na.rm = TRUE))
map(measurements, ~ mean(.x, na.rm = TRUE))
```

```
## $blood_glucose
## [1] 95
##
## $age
## [1] 30
##
## $heart_rate
## [1] 75
```

How many distinct values are in each column of `iris`:

```
map_int(iris, \(x) length(unique(x)))
```

```
## Sepal.Length  Sepal.Width Petal.Length  Petal.Width      Species
##           35           23           43           22            3
```

---

8. `map_*()` picks the type

`map()` always returns a list. The typed variants require one value of a fixed type from each element, and they return an atomic vector. A surprise length or type stops the walk, which is the point.

The suffix is the same for one input, for two, and for n.

Input 1	Input 2	Input n	Returns
`map()`	`map2()`	`pmap()`	list
`map_chr()`	`map2_chr()`	`pmap_chr()`	character vector
`map_dbl()`	`map2_dbl()`	`pmap_dbl()`	double vector (numeric)
`map_int()`	`map2_int()`	`pmap_int()`	integer vector
`map_lgl()`	`map2_lgl()`	`pmap_lgl()`	logical vector
`map_dfc()`	`map2_dfc()`	`pmap_dfc()`	data frame (by column)
`map_dfr()`	`map2_dfr()`	`pmap_dfr()`	data frame (by row)
`walk()`	`walk2()`	`pwalk()`	the input (the point is a side effect)

`map_dfc()`, `map_dfr()`, and the `map2_` and `pmap_` forms still run. purrr marks them superseded: bind the list with `list_cbind()` or `list_rbind()`.

```
map(1:2, \(i) tibble(i = i, value = i * 10)) |>
  list_rbind()

list(tibble(a = 1:2), tibble(b = 3:4)) |>
  list_cbind()
```

```
## # A tibble: 2 × 2
##       i value
##   <int> <dbl>
## 1     1    10
## 2     2    20
## # A tibble: 2 × 2
##       a     b
##   <int> <int>
## 1     1     3
## 2     2     4
```

```
map_dbl(measurements, mean)
map_chr(mtcars, class)
map_lgl(mtcars, \(x) any(is.na(x)))
map_int(iris, \(x) sum(is.na(x)))
```

```
## blood_glucose           age    heart_rate
##            95            30            75
##       mpg       cyl      disp        hp      drat        wt      qsec        vs
## "numeric" "numeric" "numeric" "numeric" "numeric" "numeric" "numeric" "numeric"
##        am      gear      carb
## "numeric" "numeric" "numeric"
##   mpg   cyl  disp    hp  drat    wt  qsec    vs    am  gear  carb
## FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## Sepal.Length  Sepal.Width Petal.Length  Petal.Width      Species
##            0            0            0            0            0
```

`class()` on an ordered factor has length 2, so `map_chr(diamonds, class)` stops at `cut`. Use `map(diamonds, class)` when a column may have more than one class.

---

9. `map2()` and `pmap()`

Count how many inputs change. The rest of the call stays fixed.

How many inputs change	Function	They line up by
1	`map()`	each element alone
2	`map2()`	position: first with first, second with second
3 or more	`pmap()`	position. Names in the list are the argument names

`map2(.x, .y, .f)` walks two objects in step. Element 1 of `.x` is paired with element 1 of `.y`.

```
map2_dbl(c(1, 2, 3), c(10, 20, 30), \(a, b) a + b)
```

```
## [1] 11 22 33
```

Step	First input	Second input	`a + b`
1	1	10	11
2	2	20	22
3	3	30	33

The same pairing draws one normal sample per mean and standard deviation. `n = 5` is fixed. `set.seed()` makes the draws match this page.

```
set.seed(14)
mu <- c(-10, 0, 10)
sigma <- c(1, 1, 2)
map2(mu, sigma, rnorm, n = 5)
```

```
## [[1]]
## [1] -10.661850  -8.281046  -7.878333  -8.502846 -10.036141
##
## [[2]]
## [1]  1.23194518 -0.06488077  1.06899373 -0.37696531  1.04318309
##
## [[3]]
## [1]  9.234356 10.598843 11.348480  9.414367 10.976107
```

`pmap()` walks a list of argument vectors. Names in the list match arguments of the function.

```
set.seed(14)
args <- list(n = 5, mean = mu, sd = sigma)
pmap(args, rnorm)
```

```
## [[1]]
## [1] -10.661850  -8.281046  -7.878333  -8.502846 -10.036141
##
## [[2]]
## [1]  1.23194518 -0.06488077  1.06899373 -0.37696531  1.04318309
##
## [[3]]
## [1]  9.234356 10.598843 11.348480  9.414367 10.976107
```

`map()` is one input. `map2()` is two. `pmap()` is the general case, including three or more.

---

10. One model per group

`group_modify()` hands each group to a function as a data frame and row-binds what you return. The function takes the group's rows and a one-row tibble of the keys. `group_map()` does the same walk and returns a list instead of one data frame.

```
mtcars |>
  group_by(cyl) |>
  group_modify(\(data, key) {
    fit <- lm(mpg ~ wt, data = data)
    tibble(term = names(coef(fit)), estimate = unname(coef(fit)))
  })
```

```
## # A tibble: 6 × 3
## # Groups:   cyl [3]
##     cyl term        estimate
##   <dbl> <chr>          <dbl>
## 1     4 (Intercept)    39.6
## 2     4 wt             -5.65
## 3     6 (Intercept)    28.4
## 4     6 wt             -2.78
## 5     8 (Intercept)    23.9
## 6     8 wt             -2.19
```

`split()` plus `map()` is the same idea without `group_by()`. A string as `.f` pulls that element out of each result. `summary(fit)$r.squared` is how `"r.squared"` finds the number.

```
mtcars |>
  split(mtcars$cyl) |>
  map(\(data) lm(mpg ~ wt, data = data)) |>
  map(summary) |>
  map_dbl("r.squared")
```

```
##         4         6         8
## 0.5086326 0.4645102 0.4229655
```

`group_walk()` is the grouped form of `walk()`: one side effect per group, nothing collected.

---

11. `walk()`, `imap()`, `map_if()`, `map_at()`

Function	What is special	Do you keep the result?
`walk()`	a file or a plot is the point	no. `.x` comes back invisibly
`group_walk()`	the same, once per group	no
`imap()`	you also receive the name, or the index	yes
`map_if()`	`.f` runs only where a test is true	yes. the other elements stay as they were
`map_at()`	`.f` runs only at the names or positions you list	yes
`pmap()`	three or more inputs at once	yes

`walk()` runs `.f` and invisibly returns `.x`. Use it when you want the plots on disk and you do not need a list of ggplot objects afterwards. This chunk is not run here. It would write three files under `figures/`.

```
dir.create("figures", showWarnings = FALSE)

variables <- c("mpg", "wt", "hp")
walk(variables, \(v) {
  p <- ggplot(mtcars, aes(x = .data[[v]])) +
    geom_histogram(bins = 15)
  ggsave(filename = paste0("figures/", v, ".png"), plot = p)
})
```

`imap()` passes the name as the second argument. `.y` in the formula form is that name.

```
imap_chr(mtcars[c("mpg", "wt")], \(x, name) paste(name, round(mean(x), 1)))
```

```
##        mpg         wt
## "mpg 20.1"   "wt 3.2"
```

`map_if()` leaves the elements that fail the test untouched. `map_at()` touches only the names you list.

```
meas <- list(a = c(1, 2, 3), b = "hi", c = c(10, 20))
map_if(meas, is.numeric, mean)
map_at(meas, "a", \(x) x * 10)
```

```
## $a
## [1] 2
##
## $b
## [1] "hi"
##
## $c
## [1] 15
## $a
## [1] 10 20 30
##
## $b
## [1] "hi"
##
## $c
## [1] 10 20
```

---

12. `map()` beside a for-loop

```
x <- list(1:3, 4:6, 7:9)
map(x, mean)

y <- vector("list", length(x))
for (i in seq_along(x)) {
  y[[i]] <- mean(x[[i]])
}
y
```

```
## [[1]]
## [1] 2
##
## [[2]]
## [1] 5
##
## [[3]]
## [1] 8
## [[1]]
## [1] 2
##
## [[2]]
## [1] 5
##
## [[3]]
## [1] 8
```

Same three means. `map()` builds the output list. The loop makes you name the index and the container.

Base R has the same jobs under other names. purrr's names say what you get back.

You want	purrr	Base R	Remember
a list	`map()`	`lapply()`	both return a list
a vector whose type you name	`map_dbl()` and the rest	`vapply()`	the type is part of the demand
two or more inputs	`map2()`, `pmap()`	`mapply()`	pair by position
a simpler vector, type not promised	you simplify yourself	`sapply()`	`sapply()` chooses the type, and it can change

A for-loop is still the right shape for `while`, for an early `break`, and for filling a structure whose next value depends on the previous one, as in the cumulative sum in section 3.

---

13. Which function

Ask down the rows. Stop at the first yes.

Ask	Function	Comes back
Inside `mutate()` or `summarize()`, same function on each column?	`across()`	columns in the data frame
One input, and a list is fine?	`map()`	a list
One input, and I need a plain vector?	`map_dbl()`, `map_int()`, `map_chr()`, `map_lgl()`	that vector. The ending is the type
Two inputs that move together?	`map2()`	a list, or `map2_dbl()` when each result is one number
Three or more inputs?	`pmap()`	a list
One result per group, stacked?	`group_modify()`	one data frame. `group_map()` is the list version
A file or a plot, and nothing to collect?	`walk()`	`.x`, invisibly. `group_walk()` is once per group
The name or the index as well?	`imap()`	a list
The function on only some elements?	`map_if()` for a test, `map_at()` for chosen names	a list

The operation is the function. The elements are `.x`. Keeping those two apart is what makes the code shorter the second time you need it.

---

14. Questions

1. What is the difference between `x[1]` and `x[[1]]` when `x` is a list?
2. Why can `map()` walk a data frame?
3. What is the difference between `map()` and `map_dbl()`?
4. How do you take the median of every numeric column?
5. How do you count missing values in every column?
6. What is the difference between `function(x)` and `\(x)`?
7. When do you use `map2()` instead of `map()`?
8. What is the difference between `group_map()` and `group_modify()`?
9. Why is `walk()` the function that saves a folder of plots?
10. How does the for-loop in section 12 become `map()`?

Answers

1. `x[1]` is a list of length 1. `x[[1]]` is the element that was inside.
2. A data frame is a list of columns. `map()` applies `.f` to each column.
3. `map()` returns a list. `map_dbl()` returns a double vector, and it stops if any element is not one double.
4. `df |> summarize(across(where(is.numeric), median))`, or `map_dbl()` after `keep(is.numeric)`.
5. `map_int(df, \(x) sum(is.na(x)))`.
6. `function(x)` and `\(x)` are the same syntax. `\(x)` is the short form from R 4.1 on. The tilde is the other form: `~` always calls the element `.x`. The table in section 7 has all four.
7. The function needs two values that change together, paired by position.
8. Both walk the groups. `group_map()` returns a list. `group_modify()` row-binds the data frames you return.
9. The point of that walk is the files. `walk()` runs the function for the side effect. `map()` would also build a list you do not need.
10. `map(x, mean)`. The index and the output list are inside `map()`.

---

15. Practice

1. The first two Fibonacci numbers are 0 and 1. Each later number is the sum of the previous two. Use a loop to compute the first 100. Sanity check: `log2()` of the 100th number is about 67.57.
2. `map_dbl()` means of every column of `mtcars`.
3. Number of unique values in each column of `iris`. Repeat with `across()`. `typeof()` of each column of `nycflights13::flights` with `map_chr()`, then with `across()`.
4. p-value of `t.test(mpg ~ am)` within each `cyl` group of `mtcars`.
5. In three lines, keep the `mtcars` columns whose mean is greater than 10 and report those means (`keep()` then `map_dbl()`). You should get four variables.
6. Standard deviation of each numeric column of `iris` with a for-loop, then with `across()`.